A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1 Negative feedback:** a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium **1 Positive feedback:** a change causes other changes that increase the original change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **Negative feedback** → a change that cancels itself out / a self-correcting cycle	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Increased low-level cloud:** higher temperatures increase evaporation and condensation, producing more cloud – clouds have a high albedo so more sunlight is reflected and warming is reduced **2 Increased photosynthesis:** higher temperatures and CO2 increase photosynthesis, removing CO2 – if stored in woody tissue atmospheric CO2 rises less **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
02.3	**1 mark each** 1 mark for each of the 4 points below. **1 Permafrost:** frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground **1** Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming **1 Methane hydrate:** methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature **1** If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **waterlogged** → soil pores filled with water so there is no air	4

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduced albedo as snow / ice melts **1** Ocean acidification reducing coral survival and carbon sequestration **1** Increased forest and peat fires releasing CO2 **1** Faster decomposition of soil organic matter releasing CO2 **1** Increased water vapour (a greenhouse gas) in warmer air **1** Increased cirrus (high) cloud which traps infrared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable **1** So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 13 / 480 = 0.0271 years **1** 0.0271 x 365 **1** = 10 days (accept 9.9-10) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (days) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1 340 000 / 425 = 3153 years (accept 3100-3200) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working.	1
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** It has a very short residence time / very rapid transfer rate (480 x 10^{15} dm^3 per year) **1** So it moves large volumes of water each year, delivering precipitation to land – small reservoirs with fast transfers respond quickly to change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** The ocean is so large that an added or removed volume is a tiny proportion of the total **1** The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
03.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) **1** But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **transpiration** → water lost as vapour from the leaves **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
04.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) **1** Very complex, expensive technology needing an advanced industrial infrastructure **1** Concerns about links between civil nuclear materials and weapons / terrorism **1** Uncertainty over permanent disposal of radioactive waste **1** Uncertain decommissioning costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Only 0.7% of natural uranium is fissile U-235 – the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) **1** Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) **1** So much more energy is harnessed from the mined uranium – fast reactors need no moderator but are complex and costly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **moderator** → slows the neutrons so that fission continues	3
04.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Advantages:** thorium is three times more abundant – much less radioactive waste with shorter half-lives – harder to make weapons material – no fuel enrichment needed **2 Disadvantages:** thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate) – U-233 is an alpha emitter so very hazardous to handle – less developed technology with high development costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
06.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Temperature:** most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster – but faster decomposition may increase deoxygenation **2 Light:** provides the activation energy for chemical reactions – photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight) – light drives photochemical smog reactions producing secondary pollutants **2 Oxygen:** needed for aerobic bacterial decomposition of organic wastes / sewage – oxidation of sulfide ores produces sulfur dioxide / acid mine drainage – oxidation state affects the solubility of metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **deoxygenation** → loss of dissolved oxygen from the water • **photochemical smog** → ground-level ozone and PANs formed by sunlight acting on oxides of nitrogen and VOCs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	6

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment **1** So they are immobilised and cannot be absorbed by organisms / leached into water – radiation is absorbed in the ground **1** But they may be released later if conditions change (eg storms disturbing sediment, low pH) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Many heavy metals (eg lead, aluminium) are more soluble at low pH **1** So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Topography:** a narrow exit from a large deep basin so a small dam holds a large volume **1 Geology:** impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity **1 Catchment area:** large enough with regular, reliable rainfall / river inflow – climate not too hot (evaporation losses) **1 Catchment land use:** low pollution risk (industry, pesticides) – low soil erosion (sedimentation reduces capacity) **1 Infrastructure / proximity to demand:** access for construction, cost of transporting water **1 Existing land use:** value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
07.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2 Advantage:** fewer land-use conflicts as no land is flooded – large volume near coastal cities **up to 2 Disadvantages:** destroys / changes important intertidal habitats (mudflats for wading birds) – obstacle to shipping – collects pollution from the whole river catchment – sedimentation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very energy-intensive and expensive (reverse osmosis / distillation) **1** Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	1 mark for the point below. **1** Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **infiltration** → water soaking into the soil surface **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) **1** A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Difference = 30 dB = 3 x 10 dB **1** = 10^3 = 1000 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** 88 dB **1** Doubling the sound intensity increases the level by 3 dB (logarithmic scale) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	1 mark for the point below. **1** 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing – quieter sounds exist but cannot be heard **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **threshold** → the concentration below which no harm is detected	1
08.5	**1 mark each** 1 mark for each of the 3 points below. **1** 20 to 20 000 Hz **1** dB(A) weights sounds between about 1000 and 4000 Hz more heavily **1** Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance – range narrows with age or noise damage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** The lowest ore purity that can be mined economically with existing technology **1** Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**1 mark each** 1 mark for each of the 6 points below. **1** X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper **1** **But X is deep:** costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) **1** X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs **1** X is far from transport infrastructure so transport costs are high **1** Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall **1** Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching) – both sulfides risk acid mine drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Higher market value / price of the metal **1** Improved extraction technology **1** Lower energy costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Smoke:** solid particulates (PM10, PM5, PM1) from incomplete combustion – small particles stay suspended, penetrate lungs – toxic chemicals (acids, phenols, metals) carried on particles – high albedo cools climate – blocks light reducing photosynthesis – damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths) – synergism with SO2 • **Controls:** Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion – smoke smogs have largely been eliminated in UK cities • **Acid rain gases:** SO2 (coal combustion, smelting sulfide ores) oxidised by ozone to SO3 - sulfuric acid – NOx (high temperature combustion) - nitric acid – HCl from PVC incineration – reactive gases travel long distances (transboundary pollution) • **Effects:** pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata) – dissolves exoskeletons – leaches Ca and Mg then mobilises toxic Al and Pb – corrodes metals, dissolves limestone – severity depends on bedrock (limestone neutralises) • **Controls:** fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays – switching from coal to gas / renewables	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Evaluation:** SO2 emissions in Europe have fallen dramatically and lake pH is recovering – NOx harder to control because of growing vehicle numbers – catalytic converters need lead-free fuel and warm engines – FGD is costly and produces wastes / gypsum – controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO2 vs NOx **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **particulates** → fine solid particles / smoke and dust • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NOx is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO_2) that caused it.	
10.2	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Reduces growth rates of trees / crops **1** Damages leaf cuticles (allowing SO2 to cause more damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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